

Cooperative binding of TEF-1 to repeated GGAATG-related consensus elements with restricted spatial separation and orientation.

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The human transcriptional enhancer factor (TEF) family includes TEF-1, TEF-3, TEF-4, and TEF-5. The TEFs share a highly conserved 68-amino acid TEA/ATTS DNA-binding domain, which binds to SV40 GT-IIC (GGAATG), SphI (AGTATG), SphII (AGCATG), and muscle-specific M-CAT (GGTATG) enhancers. We determined the optimal DNA-binding consensus sequence for TEF-1. Using a purified GST-TEF-1 fusion protein and a random pool of synthetic oligonucleotides, 31 independent clones were obtained after six rounds of binding site selection. DNA sequences analysis revealed that 16 clones contained direct repeats with a 3-bp spacer (DR3), and 15 clones contained a single binding site. The predominate consensus half-site was GGAATG (67%), and the other elements were of the form G(A)GA(T/C)ATG. The TEF-1 bound to the DR3 as a dimer in a cooperative manner. Cooperative binding was dependent on the spacing and orientation of the half-sites and was inhibited by deoxycholate treatment, providing evidence that protein-protein interactions were involved. The data suggest that TEF dimerization is important for its ability to modulate gene transcription.

Transcription factor TEAD4 regulates expression of myogenin and the unfolded protein response genes during C2C12 cell differentiation.

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The TEAD (1-4) transcription factors comprise the conserved TEA/ATTS DNA-binding domain recognising the MCAT element in the promoters of muscle-specific genes. Despite extensive genetic analysis, the function of TEAD factors in muscle differentiation has proved elusive due to redundancy among the family members. Expression of the TEA/ATTS DNA-binding domain that acts as a dominant negative repressor of TEAD factors in C2C12 myoblasts inhibits their differentiation, whereas selective shRNA knockdown of TEAD4 results in abnormal differentiation characterised by the formation of shortened myotubes. Chromatin immunoprecipitation coupled to array hybridisation shows that TEAD4 occupies 867 promoters including those of myogenic miRNAs. We show that TEAD factors directly induce Myogenin, CDKN1A and Caveolin 3 expression to promote myoblast differentiation. RNA-seq identifies a set of genes whose expression is strongly reduced upon TEAD4 knockdown among which are structural and regulatory proteins and those required for the unfolded protein response. In contrast, TEAD4 represses expression of the growth factor CTGF (connective tissue growth factor) to promote differentiation. Together these results show that TEAD factor activity is essential for normal C2C12 cell differentiation and suggest a role for TEAD4 in regulating expression of the unfolded protein response genes.

The transcription factor TEAD1 represses smooth muscle-specific gene expression by abolishing myocardin function.

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The TEAD (transcriptional enhancer activator domain) proteins share an evolutionarily conserved DNA-binding TEA domain, which binds to the MCAT cis-acting regulatory element. Previous studies have

shown that TEAD proteins are involved in regulating the expression of smooth muscle α -actin. However, it remains undetermined whether TEAD proteins play a broader role in regulating expression of other genes in vascular smooth muscle cells. In this study, we show that the expression of TEAD1 is significantly induced during smooth muscle cell phenotypic modulation and negatively correlates with smooth muscle-specific gene expression. We further demonstrate that TEAD1 plays a novel role in suppressing expression of smooth muscle-specific genes, including smooth muscle α -actin, by abolishing the promyogenic function of myocardin, a key mediator of smooth muscle differentiation. Mechanistically, we found that TEAD1 competes with myocardin for binding to serum response factor (SRF), resulting in disruption of myocardin and SRF interactions and thereby attenuating expression of smooth muscle-specific genes. This study provides the first evidence demonstrating that TEAD1 is a novel general repressor of smooth muscle-specific gene expression through interfering with myocardin binding to SRF.

Unbiased discovery of interactions at a control locus driving expression of the cancer-specific therapeutic and diagnostic target, mesothelin.

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Although significant effort is expended on identifying transcripts/proteins that are up-regulated in cancer, there are few reports on systematic elucidation of transcriptional mechanisms underlying such druggable cancer-specific targets. The mesothelin (MSLN) gene offers a promising subject, being expressed in a restricted pattern normally, yet highly overexpressed in almost one-third of human malignancies and a target of cancer immunotherapeutic trials. CanScript, a cis promoter element, appears to control MSLN cancer-specific expression; its related genomic sequences may up-regulate other cancer markers. CanScript is a 20-nt bipartite element consisting of an SP1-like motif and a consensus MCAT sequence. The latter recruits TEAD (TEA domain) family members, which are universally expressed. Exploration of the active CanScript element, especially the proteins binding to the SP1-like motif, thus could reveal cancer-specific features having diagnostic or therapeutic interest. The efficient identification of sequence-specific DNA-binding proteins at a given locus, however, has lagged in biomarker explorations. We used two orthogonal proteomics approaches--unbiased SILAC (stable isotope labeling by amino acids in cell culture)/DNA affinity-capture/mass spectrometry survey (SD-MS) and a large transcription factor protein microarray (TFM)--and functional validation to explore systematically the CanScript interactome. SD-MS produced nine candidates, and TFM, 18. The screens agreed in confirming binding by TEAD proteins and by newly identified NAB1 and NFATc. Among other identified candidates, we found functional roles for ZNF24, NAB1 and RFX1 in MSLN expression by cancer cells. Combined interactome screens yield an efficient, reproducible, sensitive, and unbiased approach to identify sequence-specific DNA-binding proteins and other participants in disease-specific DNA elements.

DNA binding of TEA/ATTS domain factors is regulated by protein kinase C phosphorylation in human choriocarcinoma cells.

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Transcription enhancer factor 1 (TEF-1) controls the expression of a diverse set of genes. Previous studies implicated protein kinase C (PKC)-mediated signal transduction in modulating TEF function. We

demonstrate that in human choriocarcinoma BeWo cells, the PKC activator 12-O-tetradecanoyl phorbol 13-acetate and PKC inhibitor bisindolylmaleimide reciprocally down- and up-regulate, respectively, TEF-mediated GGAATG core enhancer activity. In vitro TEF-1 phosphorylation with several PKC isozymes and phosphoamino acid analysis confirmed that TEF-1 is a potential PKC substrate. TEF-1.DNA complexes formed by BeWo nuclear extracts are supershifted by phosphoserine- and phosphothreonine- but not phosphotyrosine-specific antibodies, indicating that TEF-1 is phosphorylated in vivo at serine and threonine residues. The TEF-1 phosphorylation domain was localized to the third alpha-helix of the DNA binding domain and adjacent hinge region by phosphopeptide analysis. TEF-1 phosphorylation significantly reduced its DNA binding activity both in vitro and in vivo, providing a possible mechanism for the inhibitory action of PKC. Finally, BeWo cells contained abundant levels of gamma and delta PKC isoforms, and their overexpression resulted in even greater inhibition of GGAATG core enhancer activity after 12-O-tetradecanoyl phorbol 13-acetate treatment. These data strongly suggest that PKC-mediated phosphorylation is a key factor controlling TEF function.

High cancer-specific expression of mesothelin (MSLN) is attributable to an upstream enhancer containing a transcription enhancer factor dependent MCAT motif.

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Identification of genes with cancer-specific overexpression offers the potential to efficiently discover cancer-specific activities in an unbiased manner. We apply this paradigm to study mesothelin (MSLN) overexpression, a nearly ubiquitous, diagnostically and therapeutically useful characteristic of pancreatic cancer. We identified an 18-bp upstream enhancer, termed CanScript, strongly activating transcription from an otherwise weak tissue-nonspecific promoter and operating selectively in cells having aberrantly elevated cancer-specific MSLN transcription. Introducing mutations into CanScript showed two functionally distinct sites: an Sp1-like site and an MCAT element. Gel retardation and chromatin immunoprecipitation assays showed the MCAT element to be bound by transcription enhancer factor (TEF)-1 (TEAD1) in vitro and in vivo. The presence of TEF-1 was required for MSLN protein overexpression as determined by TEF-1 knockdown experiments. The cancer specificity seemed to be provided by a putative limiting cofactor of TEF-1 that could be outcompeted by exogenous TEF-1 only in a MSLN-overexpressing cell line. A CanScript concatemer offered enhanced activity. These results identify a TEF family member as a major regulator of MSLN overexpression, a fundamental characteristic of pancreatic and other cancers, perhaps due to an upstream and highly frequent aberrant cellular activity. The CanScript sequence represents a modular element for cancer-specific targeting, potentially suitable for nearly a third of human malignancies.

BIONDA: a free database for a fast information on published biomarkers.

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Because of the steadily increasing and already manually unmanageable total number of biomarker-related articles in biomedical research, there is a need for intelligent systems that extract all relevant information from biomedical texts and provide it as structured information to researchers in a user-friendly way. To address this, BIONDA was implemented as a free text mining-based online database for molecular biomarkers including genes, proteins and miRNAs and for all kinds of diseases. The

contained structured information on published biomarkers is extracted automatically from Europe PMC publication abstracts and high-quality sources like UniProt and Disease Ontology. This allows frequent content updates. BIONDA is freely accessible via a user-friendly web application at <http://bionda.mpc.ruhr-uni-bochum.de>. The current BIONDA code is available at GitHub via <https://github.com/mpc-bioinformatics/bionda>. Supplementary data are available at Bioinformatics Advances online.

Human genetics uncovers MAP3K15 as an obesity-independent therapeutic target for diabetes.

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We performed collapsing analyses on 454,796 UK Biobank (UKB) exomes to detect gene-level associations with diabetes. Recessive carriers of nonsynonymous variants in MAP3K15 were 30% less likely to develop diabetes ($P = 5.7 \times 10^{-10}$) and had lower glycosylated hemoglobin ($\beta = -0.14$ SD units, $P = 1.1 \times 10^{-24}$). These associations were independent of body mass index, suggesting protection against insulin resistance even in the setting of obesity. We replicated these findings in 96,811 Admixed Americans in the Mexico City Prospective Study ($P < 0.05$). Moreover, the protective effect of MAP3K15 variants was stronger in individuals who did not carry the Latino-enriched SLC16A11 risk haplotype ($P = 6.0 \times 10^{-4}$). Separately, we identified a Finnish-enriched MAP3K15 protein-truncating variant associated with decreased odds of both type 1 and type 2 diabetes ($P < 0.05$) in FinnGen. No adverse phenotypes were associated with protein-truncating MAP3K15 variants in the UKB, supporting this gene as a therapeutic target for diabetes.

Interactions Between Nanoparticles and Dendritic Cells: From the Perspective of Cancer Immunotherapy.

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Dendritic cells (DCs) are the primary antigen-presenting cells and play key roles in the orchestration of the innate and adaptive immune system. Targeting DCs by nanotechnology stands as a promising strategy for cancer immunotherapy. The physicochemical properties of nanoparticles (NPs) influence their interactions with DCs, thus altering the immune outcome of DCs by changing their functions in the processes of maturation, homing, antigen processing and antigen presentation. In this review, we summarize the recent progress in targeting DCs using NPs as a drug delivery carrier in cancer immunotherapy, the recognition of NPs by DCs, and the ways the physicochemical properties of NPs affect DCs' functions. Finally, the molecular pathways in DCs that are affected by NPs are also discussed.

Dendritic cells in allergic airway inflammation.

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Dendritic cells (DCs) are primary antigen-presenting cells involved in interactions with T cells leading to the proliferation of TH1 or TH2 cell types. In asthma, predominance of TH2 cells appears to be

responsible for disease pathogenesis. Differentiation of TH2 cells is driven by a variety of factors such as the expression of high levels of costimulatory molecules, the cytokine profile, and the subset of DCs. Many inflammatory cells involved in the pathogenesis of asthma either directly or indirectly modulate DC function. Traditional treatments for asthma decrease the number of airway DCs in animals as well as in patients with asthma. Immunomodulators including interleukin (IL)-10, transforming growth factor (TGF)-beta, cytosine-phosphate-guanosine-containing oligodeoxynucleotides (CpG-ODN), 1alpha,25-dihydroxyvitamin D3, and fetal liver tyrosine kinase 3 ligand (Flt3L) are involved in the modulation of the function of DCs. Based on the critical review of the interaction between DCs and other inflammatory cells, we propose that activation of T cells by DCs and sensitization to inhaled allergen and resulting airway inflammation are dependent on plasmacytoid and myeloid subset of lung DCs to induce an immune response or tolerance and are tightly regulated by T-regulatory cells. Effects of various therapeutic agents to modulate the function of lung myeloid DCs have been discussed.
